Access Control System

Cryptographic Authentication & Role-Based Authorization

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What is Access Control?

Access Control restricts who can view or use resources in a computing environment. Cryptography makes this process secure.

Authentication

Verify user identity before granting access

Authorization

Determine what an authenticated user may do

Encryption

Protect passwords using cryptographic algorithms

Analogy: A hospital where doctors, nurses, and visitors each have different levels of access — enforced by cryptography.

How Cryptography Secures Access

01

Password Hashing

Passwords are transformed using SHA-256 into a fixed hash that cannot be reversed.

02

Salt

A unique random value added to each password before hashing prevents duplicate hashes.

03

PBKDF2

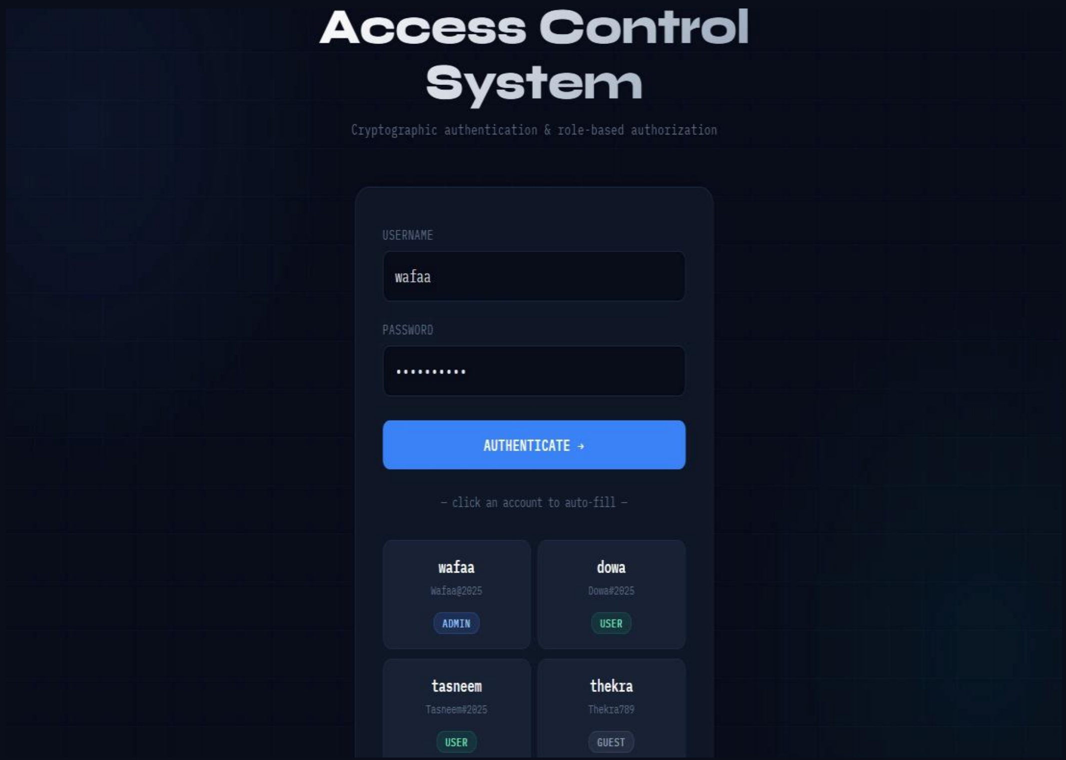
The hashing runs 100,000 iterations, making brute-force attacks extremely slow.

04

RBAC

Role-Based Access Control assigns each user a role that defines which resources they can access.

Login Page — Authentication



🔐 Secure Login Form

Username and password fields with cryptographic verification

👥 4 User Accounts

Wafaa (Admin), Dowa & Tasneem (User), Thekra (Guest)

🔑 Password Hashing

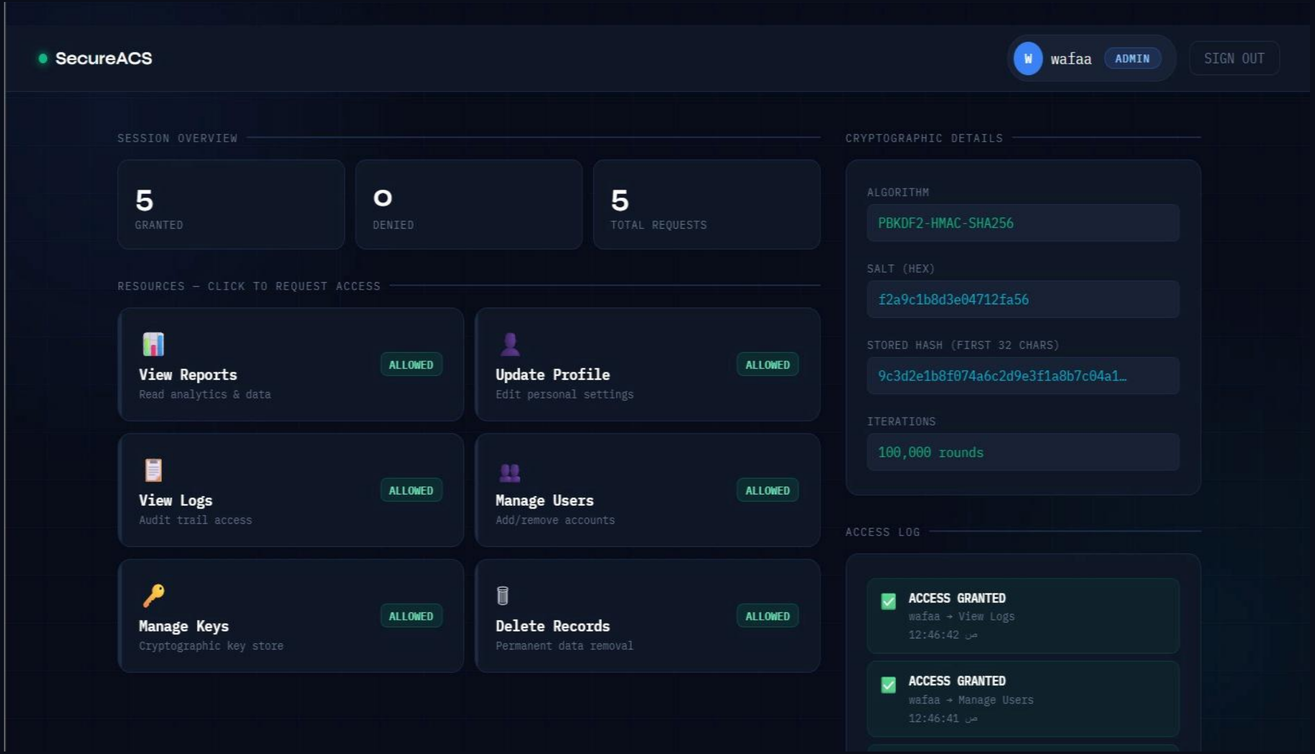
Passwords converted to SHA-256 hash — never stored as plain text

⚡ Role Assignment

After login, the system assigns the user a role determining their access

ADMIN

Wafaa — Admin Dashboard



All 6 Resources: ALLOWED

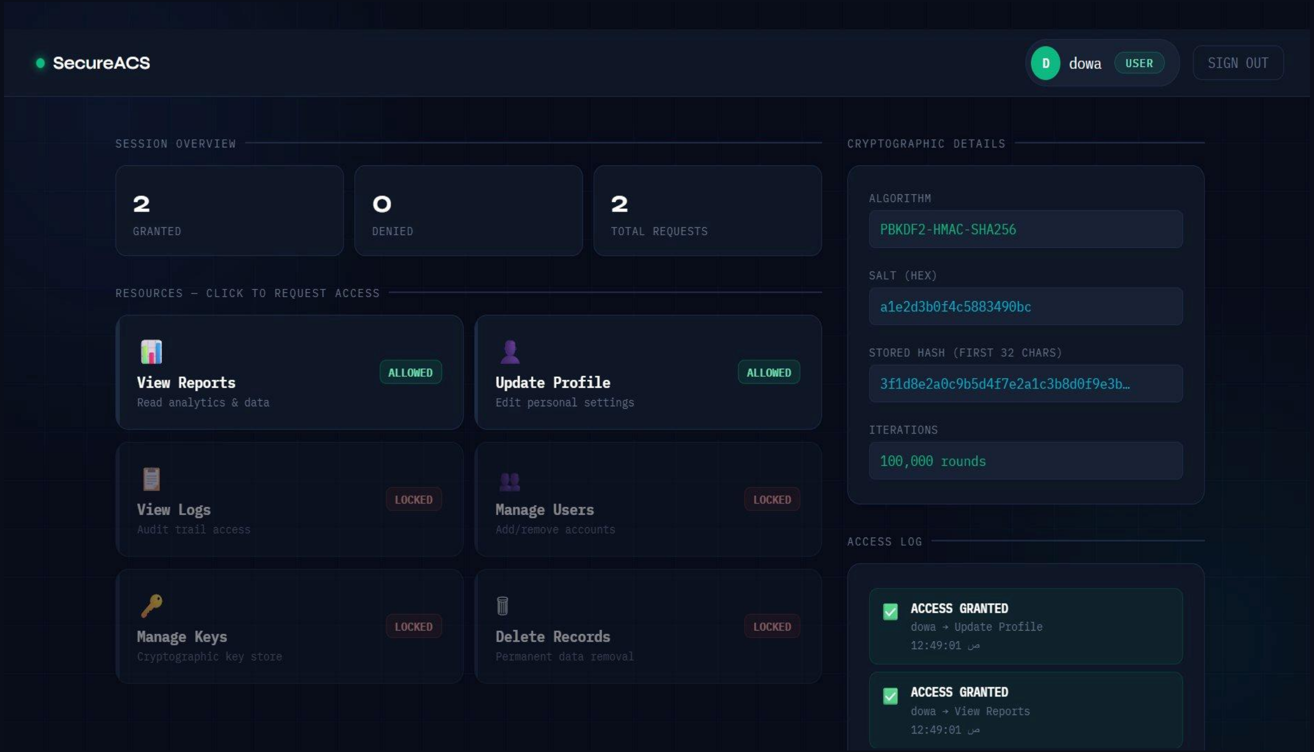
Full system access granted

Admin Permissions

- ✓ View Reports
- ✓ Update Profile
- ✓ View Logs
- ✓ Manage Users
- ✓ Manage Keys
- ✓ Delete Records

USER

Dowa — User Dashboard



2 of 6 Resources: ALLOWED

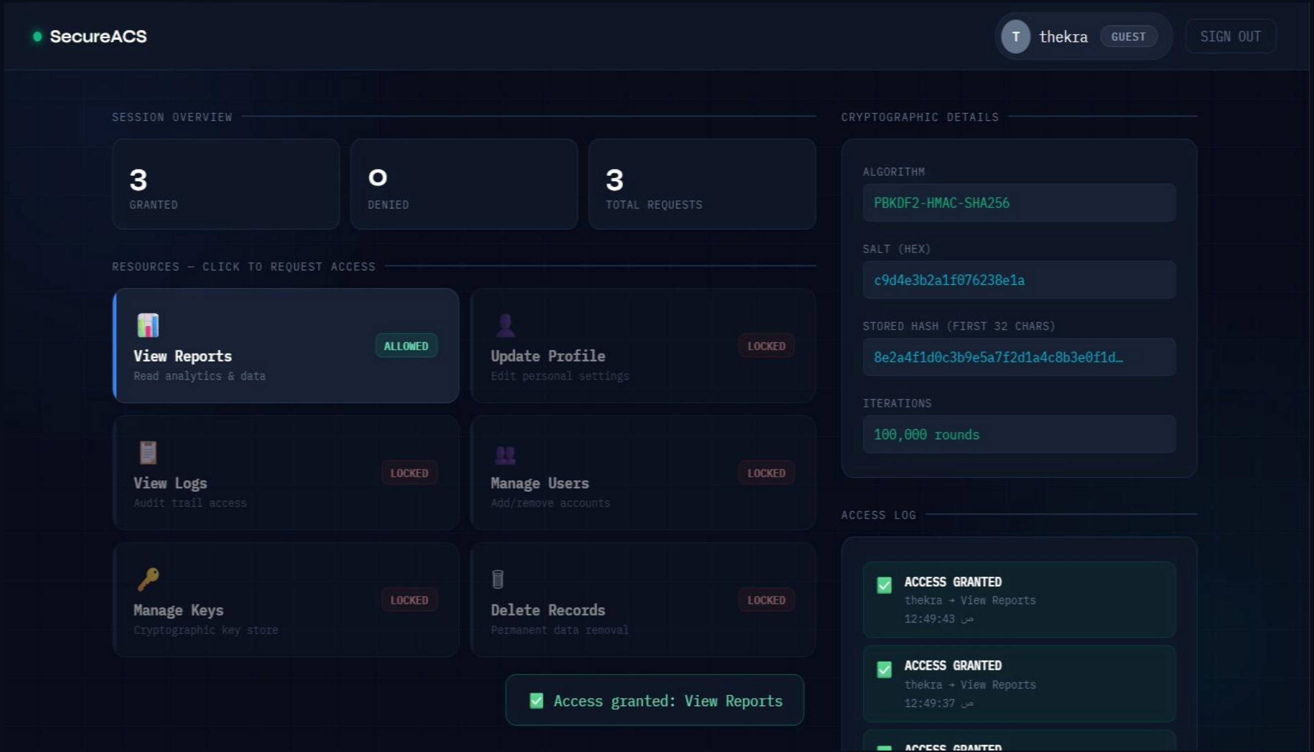
Limited access only

User Permissions

- ✓ View Reports
- ✓ Update Profile
- ✗ View Logs
- ✗ Manage Users
- ✗ Manage Keys
- ✗ Delete Records

GUEST

Thekra — Guest Dashboard



1 of 6 Resources: ALLOWED

Minimum access only

Guest Permissions

- ✓ View Reports
- ✗ Update Profile
- ✗ View Logs
- ✗ Manage Users
- ✗ Manage Keys
- ✗ Delete Records

Role Permissions Summary

Resource	Admin (Wafaa)	User (Dowa/Tasneem)	Guest (Thekra)
View Reports	✓	✓	✓
Update Profile	✓	✓	✗
View Logs	✓	✗	✗
Manage Users	✓	✗	✗
Manage Keys	✓	✗	✗
Delete Records	✓	✗	✗

How the Code Works

1. User enters password
2. System adds unique salt
3. PBKDF2-SHA256 hashes it
4. Hash compared to stored hash
5. Role assigned & access granted

hash_password()

```
salt = os.urandom(16)
key = pbkdf2_hmac(
  "sha256", password,
  salt, 100000)
```

verify_password()

```
_, attempt = hash_password(
  input, stored_salt)
return attempt == stored
```

access_resource()

```
allowed = RESOURCES[role]
if resource in allowed:
  print("GRANTED")
else: print("DENIED")
```

Real-World Applications

🏦 Banking Systems

Tellers access accounts, managers approve transfers, all cryptographically secured.

🏥 Healthcare

Doctors access records, nurses view vitals, patients see only their own data.

☁️ Cloud Platforms

AWS, Azure & Google Cloud use role-based policies for resource management.

🏛️ Government

Classified documents use cryptographic multi-level access control.

📱 Mobile Apps

OAuth and JWT are cryptographic standards used globally for app access.

🎓 Universities

Students, professors, and admins each have different access to university systems.

Key Takeaways

01

Access Control is the backbone of system security — only authorized users can reach sensitive resources.

02

Cryptography (SHA-256, PBKDF2, Salt) protects stored passwords from theft even if the database is compromised.

03

RBAC allows fine-grained permission management — each role accesses only what it needs.

04

Our SecureACS demo implements login, hashing, roles, access logs, and live enforcement in a real web interface.

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Sources & References

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[4] Ferraiolo et al. (2001). Proposed NIST standard for role-based access control. ACM TISSEC, 4(3), 224-274.

[5] Python Software Foundation — hashlib PBKDF2-HMAC docs: https://docs.python.org/3/library/hashlib.html

[6] Menezes, A. et al. (1996). Handbook of Applied Cryptography. CRC Press.